

Horizon A1–A6: From Runtime Safety Candidates to a Singapore Policy-Assurance Shadow

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ABSTRACT

Horizon places an independent runtime-assurance governor between a vessel’s decision AI and its actuators. Six architectures have been developed. A1–A5 progressively add probabilistic risk, bounded prediction, barrier filtering, and evidence-conditioned recovery to the physical-safety path. A6 is deliberately different: it is an observation-only Singapore policy shadow placed after A5. In the historical 60-branch development study, A5 achieved 99.83% gate acceptance with three deadline misses, zero declared-recoverable collisions, and zero unsafe or stale acceptances. A current-code rerun applied A6 after every A5 decision over the same 12 episode identities: all 1,788 assessments withheld support because A5 always chose recovery, minimum-risk, or invalid action; A6 median latency was 0.178 ms. The selected prototype architecture is therefore A5 for physical runtime assurance plus A6 for auditable policy shadowing. This is an engineering selection, not certification or a legal-compliance claim.

KEYWORDS

runtime assurance, autonomous vessels, maritime policy, mechanistic interpretability

1 PROTECTED SYSTEM ARCHITECTURE

The external decision AI may propose heading and speed but cannot command actuators directly. Each proposal is joined with navigation, contacts, uncertainty, platform status, communications, perception, and neural-health evidence. An assurance candidate produces a decision; an exclusive gate then revalidates identity, lineage, freshness, deadlines, actuator limits, and final-command safety. Unknown, stale, late, or unbounded evidence can only preserve or reduce authority. A watchdog retains minimum-risk authority if supervision stops.

2 A1–A6 PROGRESSION

All six architectures are deterministic engineering implementations; none uses reinforcement learning. A1–A5 share one bounded governor-input to assurance-decision contract and were compared as alternative physical-safety candidates. A6 is a complementary policy assessment and is not eligible to issue a command.

A1 is transparent and fast, but a threshold result does not establish complete trajectory safety. **A2** represents uncertainty probabilistically, but its analytic risk score is uncalibrated and is not a hard safety bound. **A3** adds finite-horizon plant rollout and demands a complete safe continuation before issuance. **A4** searches the nonlinear vessel map for a barrier-compatible correction and validates the result with a full rollout. **A5** makes that path evidence-aware: healthy evidence retains nominal bounds and a 6 m/s cap;

Table 1: Architectural progression and present disposition.

ID	Primary mechanism	Disposition
A1	CPA/TCPA thresholds and Simplex switching	baseline
A2	Analytic Gaussian collision-risk approximation	research
A3	Bounded rollout plus complete recovery continuation	retained
A4	Robust barrier correction plus 3DOF validation	retained
A5	Health-conditioned bounds, prediction, A4 filter, recovery	control lead
A6	Versioned Singapore policy proxies over matched A5 output	shadow lead

degraded required evidence expands bounds by 1.5 and caps speed at 3 m/s; invalid required evidence forces fallback. It attempts predictive recoverability, invokes A4 when needed, and falls back on incomplete, infeasible, or late results.

A6 verifies the identity and validity of the completed A5 decision, canonicalizes its inputs, checks policy-bundle hash and validity dates, uses three-valued applicability, and evaluates bounded engineering proxies for COLREG Rules 5–8 and 13–17. Its current profile is a 12 m, 40 GT, power-driven, remotely supervised Singapore government USV in daylight and clear visibility. A6 has no command, gate, plant, or actuator interface. The IMO MASS Code is design-assurance context rather than runtime law; applicable Singapore and international instruments remain controlling sources [1–3].

3 A1–A5 COMPARATIVE RESULTS

The initial development matrix contained 12 paired identities: four scenarios, three seeds, and five candidates per identity, for 60 branches. Scenarios covered recoverable crossing, dense traffic, vessel degradation, and initially unrecoverable encounters. Timing was idealized except for 20 ms candidate and gate service, so the study compares architecture behaviour rather than production latency.

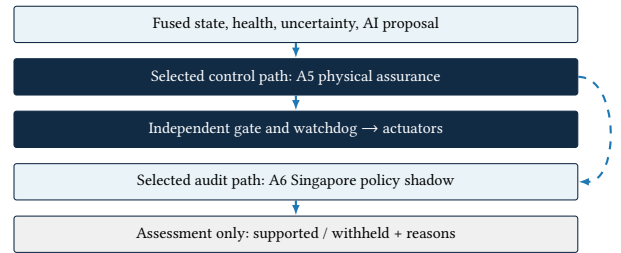


Figure 1: A5 controls the protected path; A6 can only observe it.

Table 2: A1–A5 development study; 12 branches per candidate.

ID	Gate accepted	Deadline misses	Recoverable collisions	Unsafe/stale accepted
A1	8.87%	0	0	0
A2	8.73%	1	0	0
A3	99.66%	6	0	0
A4	99.44%	10	0	0
A5	99.83%	3	0	0

Table 3: A6 shadow on the matched A5 experiment.

Measure	Result
Episodes / A6 assessments	12 / 1,788
A5 gate accepted / rejected	1,773 / 15
A5 recover / minimum-risk / invalid	888 / 885 / 15
A6 supported / withheld	0 / 1,788
A6 latency p50 / p95 / p99	0.178 / 0.281 / 0.318 ms
Collision episodes	3 initially unrecoverable

None of the 45 declared-recoverable branches collided; all 15 deliberately unrecoverable branches collided. No unsafe or stale result was accepted. A1 and A2 often passed proposals the independent predictive gate rejected, producing acceptance below 9%. A3–A5 actively selected recovery or minimum-risk commands and exceeded 99.4% acceptance. A5 had the highest acceptance and fewer deadline misses than A3 or A4, so it became the physical-control lead. These small, development-set results do not prove statistical superiority or operational safety.

4 A6 MATCHED EXPERIMENT

The A6 companion run reused the exact 12 A5 episode identities from the historical design: four scenarios, three seeds, observation and fault hashes, A1 policies, 30-second horizon, fixed health method, and idealized timing profile. A6 observed every completed A5 decision without advancing simulation time. The policy bundle, declared lookout capabilities, vessel-profile facts, and public-state evidence adapter were synthetic development fixtures; this was not legal validation.

Each scenario produced 447 A6 assessments and withheld all 447. Every assessment recorded physical A5 safety as dominant; the 15 invalid A5 decisions also recorded invalidity. The three collisions were the predeclared initially-unrecoverable cases. Compared with the historical source revision’s three deadline-invalid decisions, the current-code rerun produced 15. Because measured host work can influence A5’s deadline result, the two revision-specific counts remain separately labelled.

5 WHY THE A5+A6 COMPOSITION WAS SELECTED

There is no honest single ranking of A1–A6 because A6 solves a different problem. **A5 was selected for the prototype control path** because it combined the best historical gate acceptance with evidence-conditioned conservatism, complete recovery validation, and fewer misses than the other model-based candidates. **A6 was selected for the audit path** because it adds a missing

policy-evidence dimension without modifying A5, the gate, or actuators. It is hash-pinned, time-bounded, identity-matched, explicitly three-valued, and fail-closed: across all 1,788 matched decisions it never converted A5 recovery, minimum-risk, or invalid output into positive policy support. Its measured median local cost was 0.178 ms.

The selection is architectural, not a claim that A6 understands all maritime law. Restricted visibility and broader vessel profiles remain unsupported. Weapons, targeting, classified doctrine, and rules of engagement are excluded. Policy-evidence adapters, authoritative source ingestion, legal review, target-computer timing, sustained-load tests, and wider encounter coverage are still required.

6 NEURAL-HEALTH INPUT AND VERDICT

H5 supplies a hash-pinned, shadow-only mechanistic monitor of WaSR-T’s temporal-fusion representation. Its selected development feature beat random-direction and equal-norm causal controls in all 18 trials, but live scores remain unknown without frozen calibration. H5 therefore cannot prove obstacle-free water or increase A5/A6 authority.

The deployment verdict is: *A5 control plus A6 shadow is ready for instrumented shadow trials, not autonomous-policy authority, certification, or legal-compliance claims.* A6 should remain advisory until authoritative policy ingestion, evidence freshness and provenance, target-platform latency, and independent scenario and legal review are complete.

Reproducibility. Plan: experiment/manifests/a6-a32-shadow-development.json. Result: horizon-runs/development/a6-a32-shadow-30s-20260924-v2/analysis.json. SHA-256: f18221263d9c14fc2e16eca5c8be466b951eab60cab9d45811d49d93b2d88357.

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